SUBCHAPTER 3.5 · THERMODYNAMICS & ENERGY BALANCES

Total Energy Conservation & Thermodynamic Applications

First Law of Thermodynamics for moving control volumes, enthalpy grouping, stagnation enthalpy h_0, and engineering shaft work.

1. The Master Total Energy Conservation Equation

The First Law of Thermodynamics for a closed material fluid system $V_f(t)$ states that the rate of increase of total energy (internal energy e plus macroscopic kinetic energy $v^2/2$) equals the net mechanical power done on the fluid plus the net rate of heat transferred into the fluid:

$$\frac{d}{dt}\left[\int_{V_f(t)} \rho\left(e + \frac{|\vec{v}|^2}{2}\right) dV\right] = \dot{W}_{\text{ext}} + \dot{Q}$$

[Eq. 3.45] First Law of Thermodynamics for a closed material fluid system.

Substituting intensive total energy $b = e + \frac{|\vec{v}|^2}{2}$ into the universal Reynolds Transport Theorem yields the **Master Integral Total Energy Equation** for any moving control volume $V_c(t)$:

$$\mathbf{\frac{d}{dt}\left[\int_{V_c(t)} \rho\left(e + \frac{|\vec{v}|^2}{2}\right) dV\right] + \int_{\Sigma_c(t)} \rho\left(e + \frac{|\vec{v}|^2}{2}\right)[(\vec{v} - \vec{v}_c)\cdot\vec{n}]\, d\sigma = \dot{W}_{\text{ext}} + \dot{Q}}$$

[Eq. 3.46] Master Integral Total Energy Conservation Equation.

Here, the external power $\dot{W}_{\text{ext}}$ done on the fluid by boundary forces and body forces breaks down into:

$$\dot{W}_{\text{ext}} = -\int_{\Sigma_c} p(\vec{v}\cdot\vec{n})\, d\sigma + \int_{\Sigma_c} \vec{v}\cdot\bar{\bar{\tau}}'\cdot\vec{n}\, d\sigma + \int_{V_c} \rho\vec{f}_m\cdot\vec{v}\, dV + \dot{W}_{\text{shaft}}$$

and total thermal heat transfer $\dot{Q}$ combines boundary conduction (Fourier's law $\vec{q} = -k\nabla T$) and internal sources (chemical combustion Q_c and radiative absorption Q_r):

$$\dot{Q} = -\int_{\Sigma_c} \vec{q}\cdot\vec{n}\, d\sigma + \int_{V_c} (Q_c + Q_r)\, dV$$

Total Energy: Complete Derivation Chain (Notes.pdf, Eqs. 3.39–3.46)

Step 1: Origin of each term of the First Law (Eq. 3.45)

The total energy of a fluid volume is $\int_{V_f} \rho(e + |\vec{v}|^2/2)\, dV$. Its rate of change equals the power of the forces plus the heat added (First Law). Each contribution is built from an elementary power on a surface element $d\sigma$ or a volume element dV.

- **Surface forces.** The stress vector $(\bar{\bar{\tau}} \cdot \vec{n})\, d\sigma$ acts on material that moves with velocity $\vec{v}$, so its power is $\vec{v} \cdot \bar{\bar{\tau}} \cdot \vec{n}\, d\sigma$. With $\bar{\bar{\tau}} = -p\bar{\bar{I}} + \bar{\bar{\tau}}'$ and $\vec{v} \cdot (p\bar{\bar{I}}) \cdot \vec{n} = p\, \vec{v} \cdot \vec{n}$, the total is $-\int p\, \vec{v} \cdot \vec{n}\, d\sigma + \int \vec{v} \cdot \bar{\bar{\tau}}' \cdot \vec{n}\, d\sigma$.
- **Body force.** The force $\rho \vec{f}_m\, dV$ has power $\rho \vec{f}_m \cdot \vec{v}\, dV$.
- **Heat conduction.** By Eqs. 3.41 and 3.43, $\vec{q} \cdot \vec{n}\, d\sigma$ is the conductive heat leaving through $d\sigma$ (its divergence is the heat loss per unit volume), so the heat added is $-\int \vec{q} \cdot \vec{n}\, d\sigma$.
- **Volumetric sources.** Chemical reaction and radiation add $\int (Q_c + Q_r)\, dV$.

Summing these contributions gives Eq. 3.45:

$$\frac{d}{dt}\int_{V_f} \rho\Big(e + \frac{|\vec{v}|^2}{2}\Big) dV = -\int_{\Sigma_f} p\, \vec{v} \cdot \vec{n}\, d\sigma + \int_{\Sigma_f} \vec{v} \cdot \bar{\bar{\tau}}' \cdot \vec{n}\, d\sigma + \int_{V_f} \rho \vec{f}_m \cdot \vec{v}\, dV - \int_{\Sigma_f} \vec{q} \cdot \vec{n}\, d\sigma + \int_{V_f} (Q_c + Q_r)\, dV$$

Eq. 3.45

Step 2: Transfer to a moving control volume (Eq. 3.46)

Apply Eq. 3.8 with $\phi = \rho(e + |\vec{v}|^2/2)$ to the left-hand side of Eq. 3.45 and evaluate the right-hand side on Σ_c and V_c, which coincide with Σ_f and V_f at the instant considered. Moving the convective flux to the left gives

$$\frac{d}{dt}\int_{V_c} \rho\Big(e + \frac{|\vec{v}|^2}{2}\Big) dV + \int_{\Sigma_c} \rho\Big(e + \frac{|\vec{v}|^2}{2}\Big) [(\vec{v} - \vec{v}_c) \cdot \vec{n}]\, d\sigma = -\int_{\Sigma_c} p\, \vec{v} \cdot \vec{n}\, d\sigma + \int_{\Sigma_c} \vec{v} \cdot \bar{\bar{\tau}}' \cdot \vec{n}\, d\sigma + \int_{V_c} \rho \vec{f}_m \cdot \vec{v}\, dV - \int_{\Sigma_c} \vec{q} \cdot \vec{n}\, d\sigma + \int_{V_c} (Q_c + Q_r)\, dV$$

Eq. 3.46

Step 3: Flow work and specific enthalpy (extension)

Extension, not in the Notes. For a fixed control volume ($\vec{v}_c = \vec{0}$), move the pressure work to the left-hand side and write $p\, \vec{v} \cdot \vec{n} = (p/\rho)\, \rho\, \vec{v} \cdot \vec{n}$:

$$\int_{\Sigma_0} \Big[\rho\Big(e + \frac{|\vec{v}|^2}{2}\Big) + p\Big] \vec{v} \cdot \vec{n}\, d\sigma = \int_{\Sigma_0} \rho\Big(e + \frac{p}{\rho} + \frac{|\vec{v}|^2}{2}\Big) \vec{v} \cdot \vec{n}\, d\sigma = \int_{\Sigma_0} \rho\Big(h + \frac{|\vec{v}|^2}{2}\Big) \vec{v} \cdot \vec{n}\, d\sigma$$

with $h = e + p/\rho$. The balance for a fixed volume becomes

$$\int_{V_0} \frac{\partial}{\partial t}\Big[\rho\Big(e+\frac{|\vec{v}|^2}{2}\Big)\Big]dV + \int_{\Sigma_0} \rho\Big(h+\frac{|\vec{v}|^2}{2}\Big)\vec{v}\cdot\vec{n}\,d\sigma = \int_{\Sigma_0} \vec{v}\cdot\bar{\bar{\tau}}'\cdot\vec{n}\,d\sigma + \int_{V_0}\rho\vec{f}_m\cdot\vec{v}\,dV - \int_{\Sigma_0}\vec{q}\cdot\vec{n}\,d\sigma + \int_{V_0}(Q_c+Q_r)\,dV$$

For a moving boundary, $p\,\vec{v}\cdot\vec{n} = p\,(\vec{v}-\vec{v}_c)\cdot\vec{n} + p\,\vec{v}_c\cdot\vec{n}$. The first part joins the enthalpy flux in the same way and the second part is the work of the pressure on the moving boundary, which stays on the right-hand side.

Step 4: Gravity and the total enthalpy h_0 (extension)

Extension, not in the Notes. For gravity with z pointing upward, $\vec{f}_m = -\nabla(gz)$ (Eq. 3.14 with $U = gz$). In a steady flow, the continuity equation $\nabla\cdot(\rho\vec{v}) = 0$ (Eq. 4.8) lets the body-force power be written as a divergence:

$$\rho\vec{f}_m\cdot\vec{v} = -\rho\vec{v}\cdot\nabla(gz) = -\nabla\cdot(\rho\,gz\,\vec{v}) + gz\,\nabla\cdot(\rho\vec{v}) = -\nabla\cdot(\rho\,gz\,\vec{v})$$

Gauss' theorem converts the volume integral into $-\int_{\Sigma_0}\rho\,gz\,\vec{v}\cdot\vec{n}\,d\sigma$. Moving it to the left-hand side adds gz inside the convective flux:

$$\int_{\Sigma_0}\rho\,h_0\,\vec{v}\cdot\vec{n}\,d\sigma, \qquad h_0 = h + \frac{|\vec{v}|^2}{2} + gz$$

Step 5: Steady engineering energy equation (extension)

Extension, not in the Notes. For a steady flow in a fixed control volume, split the boundary into ports, stationary solid walls and moving blades.

- On stationary walls the no-slip condition gives $\vec{v} = \vec{0}$, so the pressure and viscous work vanish; the heat conducted through the walls defines $\dot{Q} = -\int_{\text{walls}}\vec{q}\cdot\vec{n}\,d\sigma + \int(Q_c+Q_r)\,dV$.
- On the moving blades of a machine, the (time-averaged) power of pressure and viscous forces on the fluid is defined as $\dot{W}_{\text{shaft}}$.
- At the ports, viscous work and conduction are negligible against convection (high Reynolds and Peclet numbers) and h_0, ρ, $\vec{v}$ are taken uniform, so $\int_{A_k}\rho h_0\,\vec{v}\cdot\vec{n}\,d\sigma = \dot{m}_k h_{0,k}$.

The balance of Step 4 then gives the form used in the engineering applications:

$$\sum_{\text{out}}|\dot{m}|\,h_0 - \sum_{\text{in}}|\dot{m}|\,h_0 = \dot{Q} + \dot{W}_{\text{shaft}}$$

Dimensional check: each term has units $\mathrm{kg\,s^{-1}}\cdot\mathrm{J\,kg^{-1}} = \mathrm{W}$.

2. Flow Work, Enthalpy, and Stagnation Enthalpy (h_0)

A crucial step in fluid thermodynamics is recognizing that the normal pressure work term $-\int_{\Sigma_c} p(\vec{v}\cdot\vec{n})d\sigma$ can be written as:

$$-\int_{\Sigma_c} p(\vec{v}\cdot\vec{n})\,d\sigma = -\int_{\Sigma_c} \frac{p}{\rho}\,\rho(\vec{v}\cdot\vec{n})\,d\sigma$$

Moving this "flow work" (the work done by pressure pushing fluid packets across the control surface) to the left-hand side of the energy balance groups it seamlessly with internal energy e to define **specific enthalpy** h :

$$h = e + \frac{p}{\rho}$$

Combining specific enthalpy with kinetic energy and gravitational potential energy gz defines the **Stagnation (Total) Enthalpy** h_0:

$$\mathbf{h_0 = h + \frac{|\vec{v}|^2}{2} + gz}$$

For a calorically perfect gas ($h = c_pT$), the stagnation enthalpy is directly proportional to the **stagnation temperature** T_0 (here the gz term of h_0 is neglected, which is customary for gases because $g\,\Delta z \ll c_pT$):

$$h_0 = c_pT_0 = c_pT + \frac{|\vec{v}|^2}{2}$$

The Engineering Energy Equation for 1D Streams

For steady flow through devices with discrete inlets and outlets, negligible viscous work at open ports, and steady shaft work $\dot{W}_{\text{shaft}}$ (from pumps, compressors, or turbines):

$$\mathbf{\dot{Q} + \dot{W}_{shaft} = \sum_j \dot{m}_{out,j} h_{0,out,j} - \sum_k \dot{m}_{in,k} h_{0,in,k}}$$

SPECIAL CASES IN AEROSPACE PROPULSION

- **Adiabatic Duct ($\dot{Q} = 0, \dot{W}_{\text{shaft}} = 0$):** Stagnation enthalpy is strictly conserved: $h_{0,\text{out}} = h_{0,\text{in}}$. Kinetic energy increases only at the expense of static temperature ($T_{\text{out}} < T_{\text{in}}$).
- **Combustor ($\dot{W}_{\text{shaft}} = 0$):** Combustion heat release raises stagnation enthalpy: $\Delta h_0 = q_{\text{combustion}}$.
- **Compressor ($\dot{W}_{\text{shaft}} > 0$):** Mechanical work input raises total pressure and stagnation temperature: $w_{\text{in}} = c_p(T_{0,\text{out}} - T_{0,\text{in}})$.
- **Turbine ($\dot{W}_{\text{shaft}} < 0$):** Fluid delivers shaft power to drive the engine fan/compressor: $w_{\text{out}} = c_p(T_{0,\text{in}} - T_{0,\text{out}})$.

CANONICAL WALKTHROUGH 1 · CL15 SETUP

Mixing Elbow Stream Exit Temperature T_s

Physical Setup: Two streams of an incompressible liquid (density ρ, specific heat c_p) enter an adiabatic, rigid mixing tee or elbow ($\dot{Q} = 0, \dot{W}_{\text{shaft}} = 0$). Stream 1 enters with mass flow rate $\dot{m}_1$ at temperature T_1 and speed v_1. Stream 2 enters with mass flow rate $\dot{m}_2$ at temperature T_2 and speed v_2. The mixed fluid exhausts into the atmosphere at state s through an exit nozzle of area A_s.

Objective: Determine the exit mixed temperature T_s.

1 Mass Balance

From steady continuity:

$$\dot{m}_s = \dot{m}_1 + \dot{m}_2$$

The exit discharge velocity is $v_s = \frac{\dot{m}_s}{\rho A_s}$.

2 Total Energy Balance

For an adiabatic rigid mixer without shaft work ($\dot{Q} = 0, \dot{W}_{\text{shaft}} = 0$):

$$\sum \dot{m}_{\text{in}} h_{0,\text{in}} = \dot{m}_s h_{0,s} \implies \dot{m}_1 h_{0,1} + \dot{m}_2 h_{0,2} = (\dot{m}_1 + \dot{m}_2) h_{0,s}$$

Expanding the stagnation enthalpies with $h = cT$ ($c = c_p$ for an ideal gas; c is the specific heat for a liquid):

$$\dot{m}_1\left(c_pT_1 + \frac{v_1^2}{2}\right) + \dot{m}_2\left(c_pT_2 + \frac{v_2^2}{2}\right) = (\dot{m}_1 + \dot{m}_2)\left(c_pT_s + \frac{v_s^2}{2}\right)$$

3 Solve for Exit Temperature T_s

Assumption (note): writing $h_0 = cT + v^2/2$ drops the pressure term, because for a liquid $h = e + p/\rho = cT + p/\rho$. The expression is valid for an ideal gas, where $h = c_pT$ exactly, or for a liquid only if the inlet and outlet pressures are about equal, so that the p/ρ terms cancel in the balance. In low-speed liquid flow, kinetic energy differences are negligible compared to thermal energy ($v^2/2 \ll c_pT$). The energy equation reduces to the weighted calorimetric average:

$$\mathbf{T_s = \frac{\dot{m}_1 T_1 + \dot{m}_2 T_2}{\dot{m}_1 + \dot{m}_2}}$$

When discharge kinetic energy is non-negligible (high-speed nozzle discharge; for a liquid this formula is consistent only if the pressure differences are negligible compared with hov^2, since the neglected p/ho terms are of the same order as $v^2/2$; for an ideal gas it is exact):

$$\mathbf{T_s = \frac{\dot{m}_1 T_1 + \dot{m}_2 T_2}{\dot{m}_1 + \dot{m}_2} + \frac{\dot{m}_1 v_1^2 + \dot{m}_2 v_2^2 - (\dot{m}_1 + \dot{m}_2) v_s^2}{2c_p(\dot{m}_1 + \dot{m}_2)}}$$

CANONICAL WALKTHROUGH 2 · CL10 SETUP

Ventilation Fan Nozzle Shaft Power $\dot{W}$

Physical Setup: An electric ventilation fan inside an adiabatic duct takes ambient air (ρ, p_a, T_a) from rest ($v_1 \approx 0$) and accelerates it through a contracting nozzle discharging into the room at high exit speed v_e, area A_e, at ambient pressure $p_e = p_a$. Total pressure loss due to friction across duct screens is Δp_{loss}.

Objective: Calculate the minimum mechanical shaft power $\dot{W}_{\text{shaft}}$ that must be supplied by the electric motor.

Analysis

Energy Balance from Inlet to Exit

Mass flow rate through the nozzle: $\dot{m} = \rho A_e v_e$.

Applying the engineering energy equation across the fan and nozzle ($\dot{Q} = 0$, $z_1 \approx z_e$):

$$\dot{W}_{\text{shaft}} = \dot{m}(h_{0,e} - h_{0,1}) = \dot{m}\left[\left(h_e + \frac{v_e^2}{2}\right) - (h_1 + 0)\right]$$

For an incompressible air flow ($\rho = \text{const}$), enthalpy changes arise from the internal energy generated by viscous dissipation and from pressure differences ($p_e = p_1 = p_a \implies \Delta(p/\rho) = 0$). The loss is *defined* by the dissipated energy per unit mass, $\Delta p_{\text{loss}} \equiv \rho\,(e_e - e_1)$ (adiabatic duct, so all the dissipation stays in the fluid); this is a definition, not a derived result:

$$h_e - h_1 = (e_e - e_1) + \frac{p_e - p_1}{\rho} = \frac{\Delta p_{\text{loss}}}{\rho}$$

Substituting into the shaft power expression:

$$\dot{W}_{\text{shaft}} = \dot{m}\left(\frac{v_e^2}{2} + \frac{\Delta p_{\text{loss}}}{\rho}\right) = \mathbf{Q}\left(\frac{\mathbf{1}}{\mathbf{2}}\rho\mathbf{v_e^2} + \mathbf{\Delta p}_{\text{loss}}\right)$$

where $Q = A_e v_e$ is the volumetric airflow rate (m^3/s).

PHYSICAL ENERGY BREAKDOWN

The fan shaft power is partitioned into two clear physical duties:

1. **Kinetic energy flux ($\frac{1}{2}\rho Q v_e^2$):** The rate of kinetic energy carried away by the high-speed discharge jet.

2. **Viscous dissipation power ($Q\Delta p_{\text{loss}}$):** The mechanical work converted irrecoverably into microscopic internal thermal energy due to friction across fan blades and duct screens.